

Munish Taya

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EDUCATION

Northeastern University, Boston, MA

Sep. 2016 - June 2018

Graduated MSc in Electrical and Computer Engineering,

Courses: Applied Probability and Stochastic Process, Machine Learning, Mobile and Wireless Networking, Wireless Communication, Advance Network Management, Modern Signal Processing

GPA - 3.76/4.0

Birla Institute of Technology, Mesra, India

June 2010 - June 2014

B.E in Electronics and communication Engineering

GPA - 8.15/10s

PROFESSIONAL SKILLS

Proficient in MATLAB, Embedded C, C/C++, Linux, Python, Java, Android, Latex.

EXPERIENCE

Apple, Massachusetts, USA

April 2022 – present

- Define systems design specification for 5G NR RF transmitter.
- Investigate relevant 3GPP features and formulate a feasibility study.

Analog Devices Inc., Massachusetts, USA

July 2018 – April 2022

- Designed and implemented adaptive statistical signal processing algorithms in the wide band RF Transceivers product.
- Deriving mathematical models for system calibration algorithms and simulating calibration algorithms in MATLAB.
- The algorithms focus on Quadrature Error correction, Path delay estimation and active noise cancellation with real-time tracking based on Kalman filter framework.
- Defining system-level architecture and coordinating with hardware and software teams to specify implementation details and support system-level debug.

Analog Devices Inc., Massachusetts, USA

May 2017 – December 2017

- Implemented and optimized FMCW Radar Signal Chain for Automatic cars.
- Created a Monte Carlo simulation framework for analyzing FMCW radar performance (false alarm rate, miss detection probability, etc.).

Verizon Data Services India Pvt. Ltd., Hyderabad, India

June 2014 – July 2016

- Developed JAVA web applications using Spring architecture with RESTful web service for the wireless group.

RESEARCH

Communications and Digital Signal Processing (CDSP) Center, Northeastern University, Boston MA

Jan. 2017 – June 2018

- Worked on project “Multi-FFT detection of OFDM signals in time-varying channels”.
- Implemented and analyzed the performance of Partial FFT demodulation with coherent detection for signals recorded during the 2010 Mobile Acoustic Communications Experiment.
- Implemented super resolution version of Matching Pursuit and Least squares for sparse channel estimation.

Indian Institute of Technology, Patna, India

May 2013 - July 2013

- Simulated OFDM wireless communication for various underwater channel models in MATLAB.

PROJECTS

The wireless NeuroDot sensor, Northeastern University

Oct. 2016 – April 2017

- Developed an Android app to acquire EEG data from the Teensy microcontroller using serial communication and then sending using socket communication to a server for further EEG data processing.

Track Prediction with Kalman Filter, K-Nearest Neighbors and RNN with LSTM cell, Northeastern University

Nov. 2016

- Predicted the position of the object in the next frame using the position in the previous frames.
- Kalman filter, K-Nearest Neighbors were implemented using python and Recurrent Neural Network with a Long Short-Term Memory cell with its output connected to a fully connected layer.
- LSTMs and K-NN were trained on the data set gather from KITTI and performance of each was evaluated.

PUBLICATIONS

- Taya, Munish, Amir Tadayon, and Milica Stojanovic. "Mobile Acoustic Communications: Real Data Analysis of Partial FFT Demodulation with Coherent Detection." *OCEANS 2018 MTS/IEEE Charleston*. IEEE, 2018.